EXCEL INTERVIEW QUESTIONS

Based on the Supermarket Sales Dataset

For students preparing for Data Analyst, Business Analyst, MIS Executive, Reporting Analyst, and Junior BI Analyst roles

Dataset Overview

The dataset contains 1,000 transaction records and 17 columns covering branches, cities, customer types, product lines, pricing, quantity, tax, total transaction value, payment methods, gross income, dates, and customer ratings.

Column / Field	Description
Invoice ID	Transaction identifier
Branch, City	Store location information
Customer type, Gender	Customer attributes recorded in the transaction
Product line	Product category
Unit price, Quantity	Price per unit and units purchased
Tax 5%, Total	Recorded tax and transaction total
Date, Time	Transaction date and time
Payment	Payment method
COGS, Gross income, Rating	Cost, recorded gross income, and customer rating

Assumption for formulas: The examples below assume headers are in row 1 and data is in rows 2–1001. Confirm the actual column letters in your workbook before using a formula.

Section 1: Understanding the Dataset

No.	Question	Expected answer / key points
1	What does one row represent?	One supermarket sales transaction or invoice record.
2	What are the dimensions of the dataset?	1,000 data rows and 17 columns (excluding the header row from the row count).
3	Identify numerical and categorical variables.	Numerical examples: Unit price, Quantity, Tax, Total, COGS, Gross income, Rating. Categorical examples: Branch, City, Customer type, Gender, Product line, Payment.
4	Which field can identify a transaction?	Invoice ID; verify that it is unique before relying on it as a key.
5	What is the difference between Unit price and Total?	Unit price is the price per unit; Total is the recorded transaction amount including tax.
6	What does COGS mean?	Cost of goods sold.
7	What does Gross income represent?	The gross income recorded for the transaction. In this dataset, verify its relationship with the tax field before interpreting it as profit.
8	Which fields could help analyze customer satisfaction?	Rating, Product line, Branch, City, Customer type, and Gender.
9	Which fields could help analyze payment preferences?	Payment, Total, City, Branch, and Customer type.
10	What data-quality issues would you inspect?	Missing values, duplicates, inconsistent date formats, incorrect data types, invalid values, and inconsistent category labels.

Section 2: Basic Excel Formulas

No.	Question	Example formula	Purpose
11	Calculate total sales revenue.	=SUM(J2:J1001)	Total revenue
12	Calculate average transaction value.	=AVERAGE(J2:J1001)	Average Total
13	Count the number of transactions.	=COUNTA(A2:A1001)	Transaction count
14	Calculate total units sold.	=SUM(H2:H1001)	Total quantity
15	Calculate average customer rating.	=AVERAGE(Q2:Q1001)	Average rating
16	Find the highest transaction value.	=MAX(J2:J1001)	Highest Total
17	Find the lowest transaction value.	=MIN(J2:J1001)	Lowest Total
18	Count transactions paid by Cash.	=COUNTIF(M2:M1001,"Cash")	Payment analysis
19	Calculate total sales for Branch A.	=SUMIF(B2:B1001,"A",J2:J1001)	Branch performance
20	Calculate average sales for Member customers.	=AVERAGEIF(D2:D1001,"Member",J2:J1001)	Customer-type comparison

Section 3: IF, SUMIFS, COUNTIFS, and Business Rules

No.	Question	Example formula	Purpose
21	Classify transactions above 500 as High Value; others as Regular.	=IF(J2>500,"High Value","Regular")	Transaction segmentation
22	Calculate revenue for Health and beauty.	=SUMIF(F2:F1001,"Health and beauty",J2:J1001)	Product-line performance

23	Calculate sales in Yangon for Member customers.	=SUMIFS(J2:J1001,C2:C1001,"Yangon",D2:D1001,"Member")	Multiple conditions
24	Count Female customer transactions in Branch A.	=COUNTIFS(E2:E1001,"Female",B2:B1001,"A")	Demographic analysis
25	Calculate gross income for Credit card payments.	=SUMIF(M2:M1001,"Credit card",P2:P1001)	Payment analysis
26	Count transactions with rating at least 8.	=COUNTIF(Q2:Q1001,">=8")	Customer satisfaction
27	Flag transactions with quantity of 8 or more.	=IF(H2>=8,"High Quantity","Normal Quantity")	Transaction segmentation
28	Calculate average Total for Electronic accessories.	=AVERAGEIF(F2:F1001,"Electronic accessories",J2:J1001)	Product-line comparison
29	Calculate total units sold in Branch C.	=SUMIF(B2:B1001,"C",H2:H1001)	Branch performance
30	Count transactions with Total above 1,000 and rating below 6.	=COUNTIFS(J2:J1001,">1000",Q2:Q1001,"<6")	Combined conditions

Section 4: Lookup Functions

For these examples, S2 is the cell where a student enters an Invoice ID. Confirm column letters against the original dataset.

No.	Question	Example formula	Output
31	Retrieve City for the Invoice ID in S2.	=XLOOKUP(S2,$A$2:$A$1001,$C$2:$C$1001,"Not Found")	City
32	Retrieve Payment method for the Invoice ID in S2.	=XLOOKUP(S2,$A$2:$A$1001,$M$2:$M$1001,"Not Found")	Payment
33	Retrieve Rating for the Invoice ID in S2.	=XLOOKUP(S2,$A$2:$A$1001,$Q$2:$Q$1001,"Not Found")	Rating
34	Retrieve Total for the Invoice ID in S2.	=XLOOKUP(S2,$A$2:$A$1001,$J$2:$J$1001,"Not Found")	Total
35	Retrieve Product line using INDEX-MATCH.	=INDEX($F$2:$F$1001,MATCH(S2,$A$2:$A$1001,0))	Product line

Follow-up interview question: Why might a lookup fail? Expected points include incorrect lookup values, extra spaces, mismatched data types, missing IDs, or incorrect ranges. XLOOKUP returns the first matching record, so check whether Invoice IDs are unique.

Section 5: Data Cleaning and Preparation

No.	Question	Expected answer / approach
36	How would you identify missing values?	Use filters, Go To Special → Blanks, or COUNTBLANK() for applicable columns.
37	How would you check duplicate Invoice IDs?	=COUNTIF($A$2:$A$1001,A2)>1; investigate duplicates before deleting.
38	How would you handle inconsistent date formats?	Use Power Query or Text to Columns; verify day/month interpretation before conversion.
39	How would you check whether Total equals COGS plus Tax?	=ROUND(J2-(N2+I2),2)=0; verify column letters and the dataset's business definitions first.
40	How would you check invalid ratings?	Filter or conditionally format ratings outside the documented rating scale.
41	How would you find inconsistent category names?	Use filters, distinct-value checks, a mapping table, or Power Query.
42	How would you identify unusual transaction amounts?	Use conditional formatting, quartiles, or IQR; investigate before classifying a value as an error.
43	How would you identify blank customer types?	=COUNTBLANK(D2:D1001), or filter the Customer type column for blanks.
44	Why should you not delete all duplicate rows automatically?	Repeated transactions may be legitimate; verify the key and record contents first.
45	How would you prepare data for a PivotTable?	Check headers, data types, blanks, duplicate IDs, date formats, and consistent categories.

Section 6: PivotTables and Business Reporting

No.	Question	Expected approach
46	Which city generates the highest sales?	Rows: City; Values: Sum of Total; sort descending.
47	Which product line generates the highest revenue?	Rows: Product line; Values: Sum of Total; sort descending.
48	Which payment method is used most often?	Rows: Payment; Values: Count of Invoice ID.
49	Which branch sells the highest quantity?	Rows: Branch; Values: Sum of Quantity.
50	Do Member customers spend more per transaction than Normal customers?	Rows: Customer type; Values: Average of Total.
51	Which product line has the highest average rating?	Rows: Product line; Values: Average of Rating.
52	How do sales differ by gender?	Rows: Gender; Values: Sum of Total and Count of Invoice ID.
53	How would you create a sales dashboard?	KPI cards: total sales, transaction count, average transaction value, gross income; add charts for city, branch, product line, and payment.
54	How would you let management filter by branch and customer type?	Add slicers connected to relevant PivotTables and PivotCharts.
55	How would you refresh the dashboard when data is added?	Use an Excel Table as the source and refresh PivotTables or use Power Query with Refresh All.

Section 7: Charts, Dashboards, and Advanced Excel

No.	Question	Expected answer

56	Which chart compares sales across categories?	Bar or column chart.
57	Which chart is useful for a monthly trend?	Line chart; ensure the Date field is properly formatted and grouped by month.
58	What is a KPI dashboard?	A visual summary of key business performance indicators.
59	What makes a dashboard effective?	Accurate data, relevant KPIs, clear visual hierarchy, useful filters, and readable charts.
60	What is Power Query?	A tool to import, transform, combine, and refresh data using repeatable steps.
61	What is the difference between Append and Merge in Power Query?	Append stacks rows; Merge joins tables using matching columns.
62	What is Goal Seek?	A What-If Analysis tool that finds an input value needed to achieve a specified formula result.
63	What is a macro?	A recorded or programmed sequence of actions used to automate tasks in Excel.
64	What is the difference between a normal range and an Excel Table?	Tables provide structured references, built-in filters, and expand as rows are added.
65	How would you automate a monthly sales report?	Use Power Query for preparation, PivotTables/charts for analysis, and formulas or macros when useful.

Section 8: Business Case Studies

Case Study 1: Sales Performance

Management wants to know which city and product line contribute most to sales.

- Which fields belong in the PivotTable?
- Would you use Sum or Average of Total?
- Which chart would communicate the comparison clearly?

Expected approach: Sum of Total by City and Product line; sort descending and use a bar or column chart.

Case Study 2: Customer Satisfaction

The customer experience team wants to investigate low ratings.

- How would you identify low-rated transactions?
- How would you compare ratings across product lines and branches?
- What else would you inspect before recommending changes?

Expected approach: Apply a rating threshold, summarize average rating and transaction count, and avoid assuming correlation proves causation.

Case Study 3: Payment Behavior

The finance team wants to understand customer payment preferences.

- How would you count transactions by payment method?
- How would you compare revenue by payment method?
- How would you show the results in a dashboard?

Expected approach: COUNTIF or PivotTable for counts; SUMIF or PivotTable for revenue; use a bar chart.

Case Study 4: Monthly Reporting Automation

A new sales file arrives every month, and management wants the same report updated.

- How would you combine monthly files?
- How would you standardize dates and category labels?
- How would you refresh the dashboard?

Expected approach: Use Power Query to import, clean, and append files, then refresh PivotTables and charts.

Section 9: Practical Interview Assessment

Suggested duration: 30 minutes | Total: 20 marks | Students should complete the tasks in Excel.

No.	Task	Formula / expected approach	Marks
1	Calculate total revenue.	=SUM(J2:J1001)	2
2	Calculate average transaction value.	=AVERAGE(J2:J1001)	2
3	Calculate total sales in Yangon.	=SUMIF(C2:C1001,"Yangon",J2:J1001)	2
4	Count Credit card transactions.	=COUNTIF(M2:M1001,"Credit card")	2

5	Calculate gross income for Branch A.	=SUMIF(B2:B1001,"A",P2:P1001)	2
6	Find the city for an Invoice ID entered in S2.	=XLOOKUP(S2,A2:A1001,C2:C1001,"Not Found")	2
7	Flag transactions with a rating below 5.	=IF(Q2<5,"Low Rating","Acceptable Rating")	2
8	Identify repeated Invoice IDs.	=COUNTIF($A$2:$A$1001,A2)>1	2
9	Create a PivotTable showing sales by product line.	Rows: Product line; Values: Sum of Total.	2
10	Create a chart showing transaction count by payment method.	Rows: Payment; Values: Count of Invoice ID; insert a PivotChart.	2

Important Notes for Students

- Confirm column letters and field names in the provided workbook before applying formulas; the formulas assume a common layout for this dataset.
- Use the exact category spelling found in the dataset (for example, product-line names and payment-method labels).
- Do not delete duplicates or impute missing values without checking the business context.
- The dataset contains transaction-level sales records, not customer-level history or stock inventory. It does not by itself support claims about customer retention or stock availability.
- When discussing gross income, tax, and profitability, verify the dataset definitions and business logic before interpreting the values.
- In interviews, explain the business question, why you chose a formula or PivotTable, and how you validated the result.

Suggested Class Sequence

Class	Topics	Dataset activity
1	Data understanding and cleaning	Inspect columns, missing values, duplicate IDs, and date formats
2	Formulas and conditional functions	Calculate revenue, averages, counts, and branch totals
3	Lookup functions	Retrieve transaction details using Invoice ID
4	PivotTables and PivotCharts	Analyze sales by city, branch, product line, and payment
5	Dashboard creation	Build a supermarket sales performance dashboard
6	Power Query and interview assessment	Clean, refresh, and present a repeatable report

ANSWER KEY AND WORKED SOLUTIONS

This answer key is based on the supplied 1,000-row supermarket sales CSV. Formula examples assume columns A–Q follow the field order listed in the document. If your workbook has a different layout, adjust the column letters.

Important interpretation note: in this dataset, gross income equals Tax 5%. Use the recorded field definition carefully; do not automatically interpret it as net profit.

Dataset facts

Question / Task	Answer / Formula / Result
Rows and columns	1,000 transaction rows and 17 columns, excluding the header.
Missing values	0 missing cells in total; no missing values in any of the 17 columns.
Duplicate Invoice IDs	0 duplicate IDs; all 1,000 Invoice IDs are unique.
Date range	January–March 2019. The source dates are stored as MM-DD-YY text/date values; verify the date locale when importing.

Sections 1–5 — Answers to Questions 1–45

Question / Task	Answer / Formula / Result
1. What does one row represent?	One supermarket sales transaction/invoice record.
2. What are the dimensions?	1,000 data rows and 17 columns, excluding the header row.
3. Numerical and categorical variables?	Numerical: Unit price, Quantity, Tax 5%, Total, cogs, gross margin percentage, gross income, Rating. Categorical: Branch, City, Customer type, Gender, Product line, Payment. Invoice ID is an identifier; Date and Time are temporal fields.
4. Which field identifies a transaction?	Invoice ID. It is unique for all 1,000 rows in this file.
5. Unit price vs Total?	Unit price is the price per unit; Total is the full transaction amount including the recorded tax.
6. What does COGS mean?	Cost of Goods Sold—the recorded cost associated with goods sold.
7. What does Gross income represent?	The dataset's gross income field; total = 15,379.3690. It equals Tax 5% in this file, so confirm the definition before calling it profit.
8. Fields for customer satisfaction?	Rating, Product line, Branch, City, Customer type, Gender, Total, and Quantity.
9. Fields for payment preferences?	Payment, Total, City, Branch, Customer type, and optionally Gender.
10. Data-quality checks?	Missing values, duplicate IDs, inconsistent dates/categories, incorrect data types, out-of-range ratings, invalid prices/quantities, and arithmetic consistency.
11. Calculate total sales revenue.	=SUM(J2:J1001) → 322,966.7490.
12. Calculate average transaction value.	=AVERAGE(J2:J1001) → 322.9667.
13. Count transactions.	=COUNTA(A2:A1001) → 1,000.
14. Calculate total units sold.	=SUM(H2:H1001) → 5,510 units.
15. Calculate average customer rating.	=AVERAGE(Q2:Q1001) → 6.9727 out of 10.
16. Highest transaction value.	=MAX(J2:J1001) → 1,042.6500.
17. Lowest transaction value.	=MIN(J2:J1001) → 10.6785.
18. Count Cash transactions.	=COUNTIF(M2:M1001,"Cash") → 344.
19. Total sales for Branch A.	=SUMIF(B2:B1001,"A",J2:J1001) → 106,200.3705.
20. Average sales for Member customers.	=AVERAGEIF(D2:D1001,"Member",J2:J1001) → 327.7913.
21. Classify transactions above 500.	=IF(J2>500,"High Value","Regular"). Copy down. 227 transactions exceed 500.
22. Revenue for Health and beauty.	=SUMIF(F2:F1001,"Health and beauty",J2:J1001) → 49,193.7390.
23. Sales in Yangon for Member customers.	=SUMIFS(J2:J1001,C2:C1001,"Yangon",D2:D1001,"Member") → 53,637.4755.
24. Female transactions in Branch A.	=COUNTIFS(E2:E1001,"Female",B2:B1001,"A") → 161.

25. Gross income for Credit card payments.	=SUMIF(M2:M1001,"Credit card",P2:P1001) → 4,798.4320. This is the recorded gross income field, not necessarily net profit.
26. Count ratings at least 8.	=COUNTIF(Q2:Q1001,">=8") → 329.
27. Flag quantity of 8 or more.	=IF(H2>=8,"High Quantity","Normal Quantity"). Copy down. 296 transactions have quantity ≥ 8.
28. Average Total for Electronic accessories.	=AVERAGEIF(F2:F1001,"Electronic accessories",J2:J1001) → 319.6325.
29. Total units sold in Branch C.	=SUMIF(B2:B1001,"C",H2:H1001) → 1,831 units.
30. Count Total > 1,000 and rating < 6.	=COUNTIFS(J2:J1001,">1000",Q2:Q1001,"<6") → 5.
31. Retrieve City for Invoice ID in S2.	=XLOOKUP(S2,$A$2:$A$1001,$C$2:$C$1001,"Not Found"). Enter an Invoice ID in S2; the formula returns its City.
32. Retrieve Payment method for Invoice ID in S2.	=XLOOKUP(S2,$A$2:$A$1001,$M$2:$M$1001,"Not Found").
33. Retrieve Rating for Invoice ID in S2.	=XLOOKUP(S2,$A$2:$A$1001,$Q$2:$Q$1001,"Not Found").
34. Retrieve Total for Invoice ID in S2.	=XLOOKUP(S2,$A$2:$A$1001,$J$2:$J$1001,"Not Found").
35. Retrieve Product line using INDEX-MATCH.	=INDEX($F$2:$F$1001,MATCH(S2,$A$2:$A$1001,0)). MATCH finds the row position; INDEX returns the product line.
36. Identify missing values.	Use filters for (Blanks), Go To Special > Blanks, or =COUNTBLANK(range). This dataset has no missing cells.
37. Check duplicate Invoice IDs.	=COUNTIF($A$2:$A$1001,A2)>1. TRUE means an ID occurs more than once. This file has no duplicate IDs.
38. Handle inconsistent date formats.	Use Power Query or Text to Columns; select the correct locale and confirm month/day order before conversion.
39. Check whether Total = COGS + Tax.	=ROUND(J2-(N2+I2),2)=0. Copy down; verify field definitions and column letters before interpreting results.
40. Check invalid ratings.	Filter/conditionally format values outside the documented 0–10 scale. Observed ratings are 4.0–10.0.
41. Find inconsistent category names.	Use filters, UNIQUE(), a mapping table, or Power Query to list distinct labels; standardize only confirmed variants.
42. Identify unusual transaction amounts.	Use conditional formatting, quartiles/IQR, or a boxplot. Investigate flagged values before calling them errors.
43. Identify blank customer types.	=COUNTBLANK(D2:D1001) → 0.
44. Why not delete all duplicates automatically?	A repeated row may be a valid transaction. Check the key and compare row contents before deleting confirmed duplicates.
45. Prepare data for a PivotTable.	Use a single header row, consistent types, valid dates, consistent categories, checked blanks/duplicates, and ideally convert the range to an Excel Table (Ctrl+T).

Sections 6–7 — Answers to Questions 46–65

Question / Task	Answer / Formula / Result
46. Which city generates highest sales?	PivotTable: Rows = City; Values = Sum of Total; sort descending. Naypyitaw: 110,568.7065; Yangon: 106,200.3705; Mandalay: 106,197.6720. Highest: Naypyitaw.
47. Which product line generates highest revenue?	Rows = Product line; Values = Sum of Total; sort descending. Food and beverages: 56,144.84; Sports and travel: 55,122.83; Electronic accessories: 54,337.53; Fashion accessories: 54,305.89; Home and lifestyle: 53,861.91; Health and beauty: 49,193.74. Highest: Food and beverages.
48. Most-used payment method?	Rows = Payment; Values = Count of Invoice ID. Ewallet: 345; Cash: 344; Credit card: 311. Most frequent: Ewallet.
49. Which branch sells highest quantity?	Rows = Branch; Values = Sum of Quantity. A: 1859; C: 1831; B: 1820. Highest: Branch A.
50. Member vs Normal average spend?	Rows = Customer type; Values = Average of Total. Member = 327.7913; Normal = 318.1229. Member average is higher in this dataset; this does not establish why.
51. Highest average rating by product line?	Rows = Product line; Values = Average of Rating; sort descending. Food and beverages: 7.1132; Fashion accessories: 7.0292; Health and beauty: 7.0033; Electronic accessories: 6.9247; Sports and travel: 6.9163; Home and lifestyle: 6.8375. Highest: Food and beverages.
52. How do sales differ by gender?	Rows = Gender; Values = Sum of Total and Count of Invoice ID. Female: 167,882.9250 sales across 501 transactions; Male:

	155,083.8240 sales across 499 transactions.
53. Create a sales dashboard.	KPI cards: Total sales 322,966.75; transaction count 1,000; average transaction value 322.97; recorded gross income 15,379.37. Add city, branch, product-line and payment charts.
54. Filter dashboard by branch and customer type.	Insert Branch and Customer type slicers and use Report Connections to connect them to relevant PivotTables/PivotCharts.
55. Refresh dashboard when new data is added.	Convert source data to an Excel Table, build queries/PivotTables from it, then use Data > Refresh All.
56. Chart for comparing sales across categories?	Sorted bar or column chart using Sum of Total by category.
57. Chart for monthly trend?	Line chart with Date grouped by month. Monthly sales: 2019-01: 116,291.868; 2019-02: 97,219.374; 2019-03: 109,455.507.
58. What is a KPI dashboard?	A visual summary of key measures used to monitor business performance.
59. What makes a dashboard effective?	Accurate data, well-defined KPIs, readable labels, clear chart choices, useful filters, consistent formatting, and minimal clutter.
60. What is Power Query?	Excel tool for importing, transforming, combining, and refreshing data through repeatable steps.
61. Append vs Merge?	Append stacks rows from tables; Merge joins tables side-by-side using matching key columns.
62. What is Goal Seek?	A What-If Analysis tool that changes one input value to reach a target formula result.
63. What is a macro?	A recorded or VBA-programmed sequence of actions for automating repetitive tasks.
64. Normal range vs Excel Table?	Tables have structured references, built-in filters, formula auto-fill, and expand as rows are added; normal ranges do not provide all these automatically.
65. Automate a monthly sales report.	Use Power Query to import/append files and clean data; use PivotTables/PivotCharts for reporting; add slicers/KPIs; use Refresh All and validate totals before sharing.

Section 8 — Business Case Studies: Answers

Question / Task	Answer / Formula / Result
Case 1 — PivotTable fields	Rows = City and/or Product line; Values = Sum of Total. Use a cross-tab with City in rows and Product line in columns if you need both dimensions.
Case 1 — Sum or Average of Total?	Use Sum of Total to compare total sales contribution; use Average of Total to compare typical transaction size.
Case 1 — Best chart for comparison?	A sorted bar chart or clustered column chart, with clear labels and descending sort.
Case 2 — Identify low-rated transactions.	Add =IF(Q2<5,"Low Rating","Other") and filter Low Rating. There are 153 transactions below 5.
Case 2 — Compare ratings by product line and branch.	PivotTable: Rows = Product line or Branch; Values = Average of Rating and Count of Invoice ID. Use slicers if useful.
Case 2 — What else inspect?	Compare sample sizes, rating distributions, product mix, branch, date/time and other available context. Association alone does not prove causation.
Case 3 — Count transactions by payment method.	Use COUNTIF for each payment label or a PivotTable with Payment in Rows and Count of Invoice ID in Values. Ewallet: 345; Cash: 344; Credit card: 311.
Case 3 — Compare revenue by payment method.	Use SUMIF on Payment and Total or a PivotTable with Payment in Rows and Sum of Total in Values.
Case 3 — Dashboard presentation.	Show a bar chart for transaction count and a second chart for Sum of Total by payment; include total revenue and transaction count KPI cards.
Case 4 — Combine monthly files.	Data > Get Data > From Folder; combine files, check column names and types, and load the result.
Case 4 — Standardize dates/categories.	Set the correct date locale/type in Power Query, trim/clean text, and map only confirmed category variants to a standard label.
Case 4 — Refresh dashboard.	Add the new file to the source folder, select Data > Refresh All, then validate row counts, date coverage, and totals.

Section 9 — Practical Interview Assessment: Answer Key

Question / Task	Answer / Formula / Result
1. Calculate total revenue.	=SUM(J2:J1001) → 322,966.7490.
2. Calculate average transaction value.	=AVERAGE(J2:J1001) → 322.9667.
3. Calculate total sales in Yangon.	=SUMIF(C2:C1001,"Yangon",J2:J1001) → 106,200.3705.
4. Count Credit card transactions.	=COUNTIF(M2:M1001,"Credit card") → 311.
5. Gross income for Branch A.	=SUMIF(B2:B1001,"A",P2:P1001) → 5,057.1605.
6. Lookup city for Invoice ID in S2.	=XLOOKUP(S2,A2:A1001,C2:C1001,"Not Found"). The output depends on the ID entered in S2.
7. Flag rating below 5.	=IF(Q2<5,"Low Rating","Acceptable Rating"). Copy down; 153 rows have rating below 5.
8. Identify repeated Invoice IDs.	=COUNTIF($A$2:$A$1001,A2)>1. This dataset has no repeated Invoice IDs.
9. PivotTable sales by product line.	Rows = Product line; Values = Sum of Total. Highest: Food and beverages = 56,144.8440.
10. Chart transaction count by payment method.	Rows = Payment; Values = Count of Invoice ID; Insert PivotChart. Ewallet: 345; Cash: 344; Credit card: 311.

Formula and Interpretation Reminders

- Use exact category spellings from the source file, including 'Ewallet', 'Credit card', 'Health and beauty', and 'Electronic accessories'.
- For transaction-by-transaction classifications, enter the formula in row 2 and fill down.
- Use Sum of Total for total sales contribution, Average of Total for average transaction value, Count of Invoice ID for transaction counts, and Sum of Quantity for units sold.
- The data is transaction-level; it does not independently support customer retention or inventory availability analysis.
- Gross income equals Tax 5% in this dataset. Do not automatically interpret the gross income field as net profit.